Jonathan Gaynor

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EDUCATION

Columbia University | New York, NY | *M.S. in Mechanical Engineering* | Expected Graduation: December 2027

- **Fall 2026 Core Coursework:** Finite Element Analysis (ANSYS frameworks), Data Science for Mechanical Systems, Solid Biomechanics, Mechatronics and Embedded Microcomputer Control.

Grinnell College | Grinnell, IA | *Bachelor of Arts in General Science (Concentration in Mathematics)* | August 2024 – May 2026

Penn State University | Lemont Furnace, PA | *Coursework in Electrical-Mechanical Engineering Technology* | August 2022 – May 2024

TECHNICAL SKILLS

- **Design & Simulation:** SolidWorks, Fusion 360, FreeCAD, AutoCAD, Tinkercad.
- **Embedded Systems & Coding:** C/C++, Python (NumPy, Pandas), MATLAB, Arduino IDE, Java, RStudio, Jupyter Notebook.
- **Fabrication & Additive Mfg:** SLA/FDM 3D Printing, Advanced Toolpath Slicing (PrusaSlicer, Ultimaker Cura, ideaMaker), Multi-Material Printing, Mechanical Assembly.

ENGINEERING & RESEARCH PROJECTS

Catheter Instrumentation Machine | LaunchBox Grant | *Lead Systems Developer* | June 2024

- **Engineered a low-cost medical validation apparatus** integrating an automated load-cell force measurement system and thermal instrumentation arrays.
- **Programmed responsive embedded feedback loops** in C++ to process multi-sensor data streams and execute precision testing intervals.
- **Reverse-engineered commercial micro-catheter layouts** to optimize structural compactness, reducing system fabrication costs from over $1,000 to under $25.

Multi-Joint Mechatronic Robotic Arm | *Mechanical Design Lead* | November 2023 – December 2023

- **Synthesized multi-degree-of-freedom robotic kinematics layouts** in SolidWorks, validating link dimensions via continuous geometric motion testing.
- **Fabricated 3D-printed structural components in PLA**, using empirical failure analysis to optimize internal infill densities and center-of-mass balancing.
- **Integrated high-torque actuator arrays** with microprocessor control boards to synchronize mechanical, electrical, and control loops.

Autonomous Line-Following Vehicle | *Hardware Co-Designer* | April 2024

- **Integrated dual DC actuators** to establish vehicle propulsion, balancing wheel positioning for continuous motion.
- **Optimized physical layout tolerances** for optoelectronic sensor arrays, refining LED spacing to improve error-correction tracking margins.

WORK & VOLUNTEER EXPERIENCE

Helping Hands | *Mechanical Engineering Volunteer* | August 2020 – Present

- **Designed and manufactured 3D-printed prosthetic arms for underprivileged children**, using CAD software, PrusaSlicer, and FDM 3D printers to model, print, and assemble 20+ precision components per arm for functional, custom-fit solutions.
- **Produced and donated 3D-printed PPE during the COVID-19 pandemic**, designing face shields with CAD, overseeing end-to-end fabrication, and delivering dozens of completed units to Jackson Memorial Hospital in Miami, FL.

Gaynor & Company Inc. | *IT Advisor and Customer Service Assistant* | June 2022 – Present

- **Provided IT advisory support and rapid technical troubleshooting**, diagnosing and resolving a wide range of hardware and software issues to minimize business disruption and maintain operational efficiency.
- Increased policy renewals through clear risk–benefit communication and accurate document handling.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **NCAA Collegiate Baseball**: Left-Handed Pitcher & Team Leader (Grinnell College & Penn State University).
- **Maker's & 3D Printing Clubs**: Active Member, specialized in advanced slicing toolpaths and multi-material arrays.
- **Aeronautic Learning & Engineers Club**: Collaborated on student-led aerodynamic structural concepts.
- **Hobbies & Interests**: Jazz Band (Piano), South Beach Track Club, DND Club.